Composition Without Grammar: Next-Token Models Miss Regulatory Structure in Bacterial DNA

Marc Augustine Sojerido Amil
King's College London
marc.amil@kcl.ac.uk

Aleksej Zelezniak
King's College London
and Chalmers University of Technology
aleksej.zelezniak@kcl.ac.uk
aleksej.zelezniak@chalmers.se

Abstract

Natural selection jointly shapes bacterial genes and their cis-regulatory contexts. Context-aware regulatory-sequence design therefore requires more than generating species-typical DNA: upstream regions must encode valid positional grammar and, where relevant, be compatible with the target coding sequence. Because genomic language models are trained on complete genomes, one might expect them to capture these dependencies. We assess whether these models capture local, position-dependent promoter grammar and statistical associations between coding sequence (CDS) and its upstream region. In a comparative analysis of singleton orthologues, upstream similarity increased with CDS relatedness, whereas matched non-homologous controls showed little association. Motivated by this observation, we evaluate three tasks in *E. coli* and *B. subtilis*: promoter-element placement, CDS-conditioned upstream generation, and retrieval of the native upstream among 99 decoys. Across five autoregressive models, promoter placement was comparable to shuffled controls. After fine-tuning Evo with a species token, generated sequences reflected the base composition of the requested species and included a Shine–Dalgarno signal. However, replacing, shuffling, or removing the CDS produced only small changes in the generated upstream. Lastly, in the retrieval task, the best autoregressive model ranked the native upstream first in 6.3% of cases. By contrast, the contrastively trained C3P model reached 82.7% on *E. coli*, which was excluded from its training set. Under the evaluated settings, next-token models therefore capture broad composition more readily than motif identity, spacing, or gene-specific association, whereas a contrastively trained model can exploit the pairwise signal.

1 Introduction

Regulatory regions and the genes they control are shaped by related evolutionary pressures. Previous work has identified coordinated evolutionary patterns between coding sequences and their upstream regulatory contexts, reflecting selection associated with gene function, expression requirements, cellular cost and the regulatory machinery of the host [20, 21]. Near the translation start site, this relationship is further shaped by direct sequence-level constraints involving RNA structure, codon choice and translation initiation [3, 7, 18]. These associations are also apparent in a simple comparison of orthologous genes (Fig. 1A; Appendix C). Across $22{,}393$ pairs of orthologues representing monocistronic genes from 408 gammaproteobacterial genomes, nucleotide identity at third codon positions—a measure of CDS relatedness that is less constrained by amino-acid conservation—was positively correlated with the k-mer similarity of the corresponding 200 bp upstream regions (Spearman's $\rho = 0.437$, $p < 10^{-300}$). Length-matched pairs of non-homologous monocistronic genes

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from the same genomes showed little correlation ($\rho = 0.027$, $p = 4.7 \times 10^{-5}$), suggesting that broad differences in GC content, codon usage and promoter composition alone do not explain this pattern. More closely related coding sequences therefore tend to be accompanied by more similar upstream regions, raising the question of whether genomic language models learn this association.

Genomic language models are trained on complete genomes containing coding and regulatory sequence and could, in principle, learn this association. They are increasingly used to generate sequences ranging from regulatory elements to complete CRISPR–Cas systems and genome-scale DNA [2, 13, 19]. We ask two questions in this work. Local promoter grammar asks whether a model learns motif identity and positional relationships such as distal–proximal box spacing. Gene conditioning asks whether it understands the longer-range association between an upstream region and a particular CDS, rather than producing a generic species-typical sequence.

These capabilities may be challenging for next-token prediction as a bacterial upstream contains a few short, identifiable elements—promoter boxes, the Shine–Dalgarno sequence and binding sites—within a larger spacer background [6, 16–18], and cross-entropy weights every position equally (§6). Related evidence comes from perturbing natural sequences and evaluating their likelihoods [4]. Here we instead ask the models to write upstream sequence, also testing long-range dependence between the CDS and its upstream region.

This paper. Motivated by Fig. 1A, we ask whether autoregressive DNA models write upstream sequences with the annotated local elements and whether their predictions depend on the particular CDS. Three tasks (Fig. 1B–D) test this: promoter-grammar completion (§3), CDS-conditioned generation from a fine-tuned Evo [13] (§4), and recognition of the true upstream among 99 same-genome decoys (§5). After evaluation, we find limited evidence for either capability: the models reproduce broad composition but show little evidence of consistent promoter-element placement or sensitivity to the supplied CDS. A model trained by contrastive loss shows that the pairwise signal between CDS and upstream can be recovered with a different objective.

2 Problem, hypotheses and evaluation design

Biological task and data. The target is not exact reconstruction of one native upstream sequence, as generating the specific 200 bp sequence base for base is a hard problem. Instead, the evaluation measures two dependencies: local element identity and spacing, and a distributional response to the gene being regulated. The *upstream-only* track contains 671 *E. coli* K-12 MG1655 genes with up to 200 bp upstream, RegulonDB annotations [17], and assignments to one of five sufficiently represented sigma factors (Appendix D). Since *E. coli* K-12 MG1655 occurs in several pretraining corpora, each prompt has a *leaked* version containing the natural sequence and a *clean* version that retains all annotated elements, such as the proximal and distal box, at their native positions while shuffling the other bases. The *CDS-prompted* track contains every qualifying singleton gene with a complete 200 bp upstream in *E. coli* MG1655 (1,885) and *B. subtilis* 168 (1,272; Appendix F). The retrieval track evaluates 500 *E. coli* singletons, whose upstream reaches a full 200 bp before the neighbouring gene—each against 99 same-genome decoys over 20 resamples.

Models and evaluation approach. The upstream-only track uses five published autoregressive DNA checkpoints with their weights unchanged: Evo 1 at both released context lengths, 8k and 131k (`evo-1-8k-base`, `evo-1-131k-base`) [13], Evo 2 (`evo2_1b_base`) [2], HyenaDNA (`hyenadna-medium-450k-seqlen`) [12] and PromoGen2 (`promogen2-base`) [19]. All but HyenaDNA, which was trained on human hg38, saw bacterial genomes in pretraining. The CDS-prompted track evaluates fine-tuned `evo-1-8k-base`. Recognition additionally uses frozen C3P (100M), a contrastive promoter–protein model pretrained on 88 million bacterial pairs with *E. coli* held out [5]; it is used for retrieval and not as a generator. Every generative prompt is decoded 20 times at $T = 1.0$.

Hypotheses and failure criteria. Under **H1**, a model has learned local grammar if it places a same-sigma-factor proximal box within ± 2 bp of the annotation more often than composition-preserving controls, and recovers its bases at the annotated position (*placement* and *motif identity*, Appendix A). The ± 2 bp window is a loose reading of a tight biological constraint: in real σ^{70} promoters the distal–proximal spacer is sharply peaked, sitting in the canonical 15–19 bp window 60.5% of the

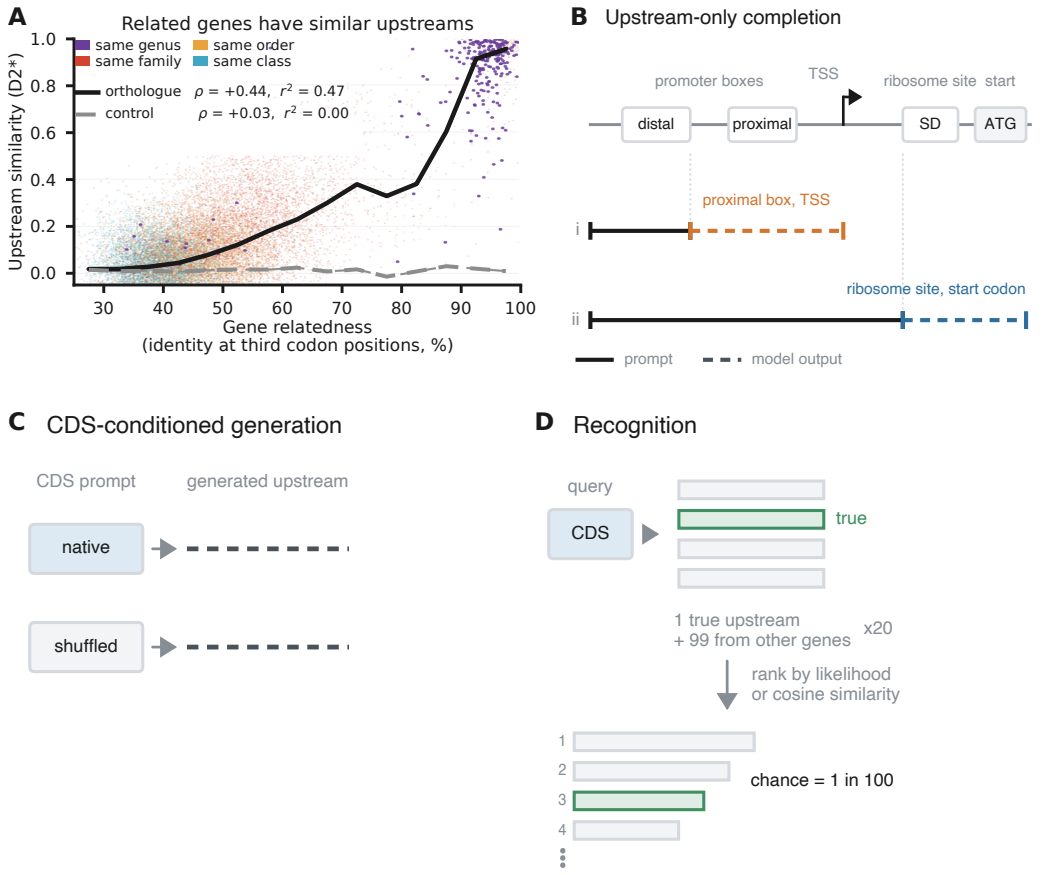


Figure 1: **Gene–upstream association and the three evaluation tasks.** **(A)** Each point is one *E. coli* K-12 singleton gene paired with a singleton orthologue in another Gammaproteobacterial genome (22,393 pairs, 408 genomes; both genes have an operon size of one and at least 200 bp to the nearest upstream feature, so every window is intergenic): identity at third codon positions against alignment-free similarity (D_2^*, 4-mers against an order-2 Markov background) of the two 200 bp upstream regions. Colour is the target genome's taxonomic rank relative to *E. coli*; curves are running medians for orthologues (black) and for a length-matched non-homologous singleton from the same genome (grey dashed). **(B)** *Upstream-only completion* (§3). A map of the upstream region with two prompt cut points: in (i) the prompt ends at the distal box and the model writes the proximal box and the transcription start site (TSS, orange); in (ii) it ends before the Shine–Dalgarno and the model writes the SD and the start codon (blue). Solid lines mark the prompt, dashed lines what the model generates. **(C)** *CDS-conditioned generation* (§4). The model generates the upstream from the native coding sequence and repeated with the CDS composition shuffled. **(D)** *Recognition* (§5). For a query CDS, the true upstream (green) is ranked against 99 upstreams of other genes (grey), by likelihood or by embedding cosine similarity, over 20 decoy sets per gene; chance is 1-in-100.

time (Fig. D.3). Under **H2**, a conditional generator uses the gene if the upstream it writes changes when the prompted CDS changes. We generate under two prompt conditions, the native CDS and a shuffled version of that same CDS, and compare the two sets of upstreams by KL divergence (Appendix A). Retrieval checks the same dependence without generating anything: top-1 accuracy should exceed the 1-in-100 chance rate when all decoys come from the same genome.

CDS-conditioned fine-tuning. We fine-tuned Evo on bacterial genes with the coding sequence first and the upstream after it—`+#CDS+$upstream` for *E. coli* and `@#CDS@$upstream` for *B. subtilis*. The objective is plain next-token cross-entropy with no masking.`+` and `@` correspond to the species token to generate species-specific sequences. The CDS→upstream order reverses natural left-to-

right genome modelling but is required for causal conditioning: an autoregressive model predicts only later tokens, so the CDS must precede every upstream token. This conditional design follows control-token generators [8, 9, 11, 19]. It also matches the practical case: a user typically has a protein they want expressed and needs regulatory sequence to drive it, so the CDS is the given and the upstream is what the model must supply.

Retrieval scoring. Autoregressive models are evaluated using candidate-centred conditional log likelihood and centred cosine similarity between CDS and upstream embeddings. C3P instead uses protein–upstream cosine similarity (Appendix A). Rank 1 places the native upstream above all 99 decoys.

3 Observed outcome I: models miss promoter-element placement

The first task tests whether the models can generate a real upstream sequence given the prompt of the end of the distal box. It must write the proximal box with the correct spacing, which is tightly constrained. We locate the box by taking the highest-scoring position-weight-matrix (PWM) window of the proximal box in the generation and count it correct if it falls within ± 2 bp of the annotated position; the same high scoring PWM is also used to assess base comparisons to the true bases of the proximal box scored in Fig. 2C (Appendix A).

No tested checkpoint writes or places the proximal box correctly. All models place the proximal box near the shuffled baseline (Fig. 2A; per-sigma-factor values in Table A.2), whereas the identical scan recovers the annotated position in 68% of real promoters. For four of five checkpoints, scrambling the generated output gives an equal or higher score, so base order adds little information about box position. The space between distal and proximal box distribution for real promoters is sharply peaked at 17 bp, whereas the distribution from every model is nearly flat across the scanned window (Fig. D.3). Accordingly, a 17 bp spacer occurs in 21.0% of real promoters but only 2.0–3.0% of model generations, close to the 2.5% expected from uniform placement. The models are therefore not simply missing the annotated position by a few bases; they fail to capture the narrow spacer-length constraint of real σ^{70} promoters. Access to a potentially memorised genomic background does not rescue performance: Evo 1 and Evo 2 saw *E. coli* during pretraining, yet shuffling non-regulatory prompt bases while preserving annotated elements changes placement by at most 1.6 percentage points and by 0.3 points for Evo 1 (8k). The bases are also incorrect. We take the highest-scoring proximal-box PWM window anywhere in the generated 45 bp and compare its six bases with that gene's annotated box. The same scan on the real continuation scores $4.99/6$. Every checkpoint scores 2.96–$3.62/6$, inside the 2.94–3.65 range of its own shuffled output (Fig. 2C). So the best box a model writes is no closer to the real one than a shuffle of its own output. The Shine–Dalgarno behaves the same way: every model scores 2.6–$3.0/6$ against $5.01/6$ for real sequence (Fig. A.2).

The models fit composition, not position. Over the 41 bp following the distal box, the four base probabilities remain flat: entropy changes at most 0.03 bits for four of the five models (Fig. 2B). Real promoters do move, falling to 1.72 bits inside the proximal box where A and T reach 0.47 and 0.45. What the models get right is the average level: their mean $P(\mathrm{A}) + P(\mathrm{T})$ is 0.57–0.61 against 0.62 in real upstreams, PromoGen2 aside at 0.78. This reflects the training dataset of the models being AT-rich in the non-coding DNA (Appendix E).

The flatness is specific to regulatory sequence. Prompted at the midpoint of a coding sequence instead, the same models write a profile that fluctuates from position to position and follows a real CDS profile: Evo 2 and Evo 1 reach 0.21 and 0.88 of a composition-only reference, against 1.05 and 1.16 on the upstream, where 1 means composition and nothing more (Appendix B). Coding sequence constrains most positions through reading frame, codon periodicity and codon usage.

More context does not help. The 200 bp window is not the limit: extending the prompt to 300, 500 and 1000 bp, and on to each checkpoint's trained limit — up to 131 kb for `evo-1-131k-base` and 450 kb for HyenaDNA — leaves three of four models at chance and lifts Evo 2 only to $1.29\times$ chance (Fig. 2D). Beyond 200 bp there is little regulatory sequence as the median *E. coli* gene has only 203 bp of intergenic sequence in front of it, so a 1000 bp prompt runs back into the neighbouring gene for 98.7% of genes; the extra context is mostly more coding sequence, not more regulation.

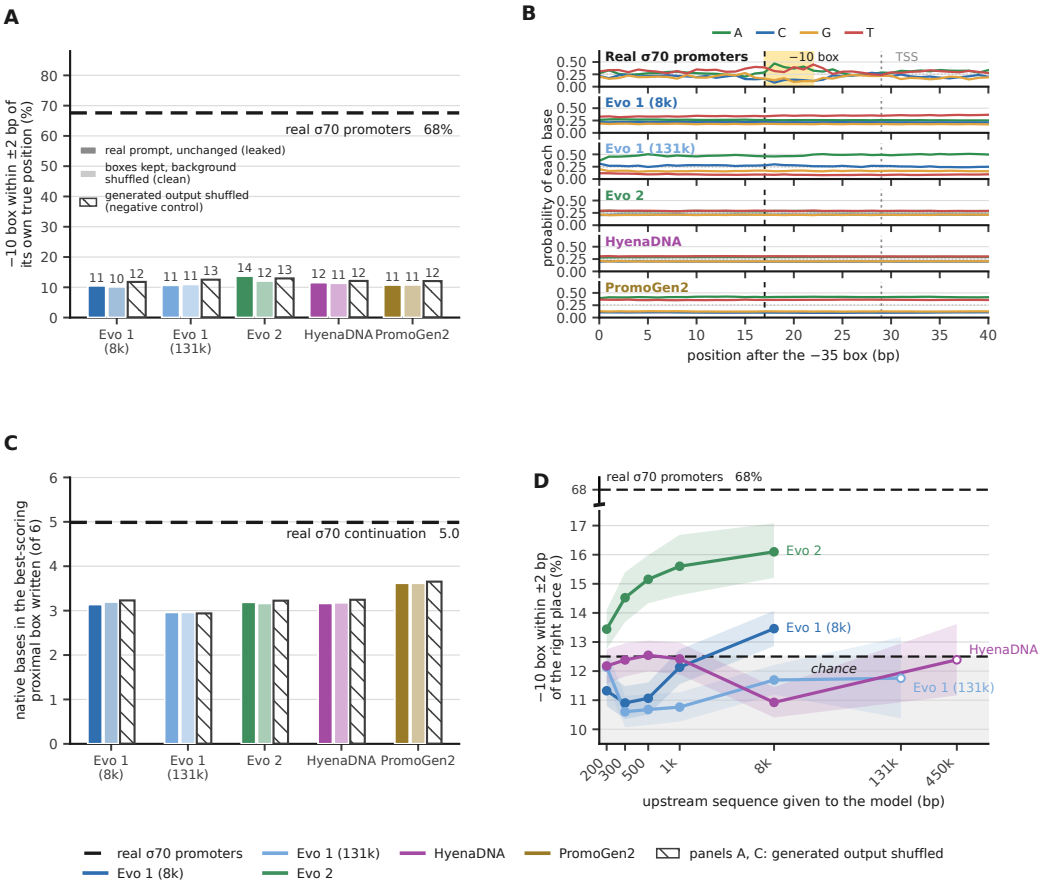


Figure 2: **Promoter-element placement after upstream completion.** Five checkpoints continued *E. coli* σ^{70} upstreams from the end of the distal box (20 generations per promoter, $T = 1.0$). **(A)** Does the proximal box land in the right place? Fraction with a proximal-box PWM match within ± 2 bp of the annotation, for natural, clean and shuffled-output conditions. **(B)** Real sequence has a signal at the proximal box that model cannot capture. Mean base probabilities over the 41 bp following the distal box, which spans the proximal box and the TSS. **(C)** Do the highest-PWM box bases match the true proximal-box bases? Mean bases of the highest-scoring proximal-box PWM window in each generation that match the gene's annotated box. **(D)** More context does not fix it. Placement accuracy as prompt context increases. Shading gives 95% confidence intervals.

4 Observed outcome II: fine-tuning does not make next-token models CDS-sensitive

Since the base autoregressive models performed poorly on promoter placement, we fine-tuned Evo to test whether conditional training can recover both species-level sequence and the gene–upstream association. During fine-tuning, the model receives a species token followed by the coding sequence, then the corresponding 200 bp upstream region. Each example follows the format `<sp>#CDS<sp>$upstream`, where `<sp>` identifies the species, `#` marks the CDS, and `$` marks the start of the upstream sequence. The experiment asks whether its distribution changes under controlled CDS perturbations. For each gene, we generate 20 sequences at $T = 1.0$ under the seven conditions in Table F.1, which differ only in the CDS. To separate the learned species prior from CDS dependence, Fig. 3A–B first omits the CDS entirely and Fig. 3C–D then analyzes CDS-prompted outputs.

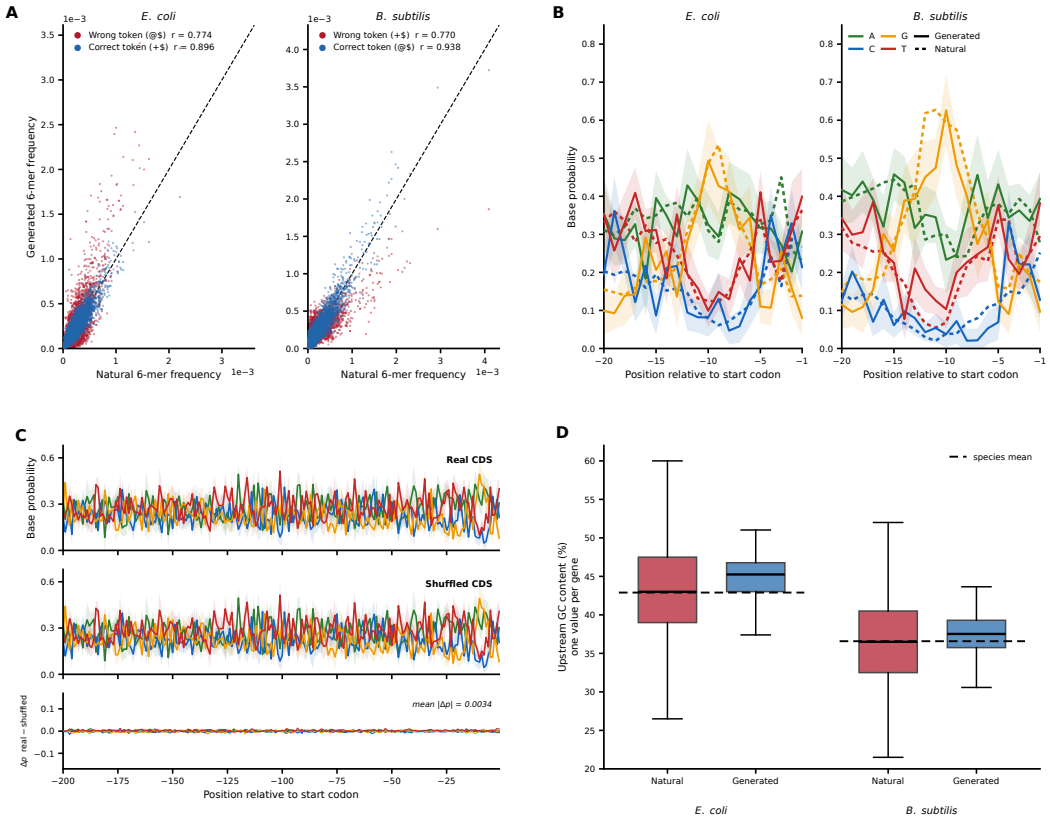


Figure 3: **Generated upstream properties and sensitivity to CDS perturbations.** All outputs use the fine-tuned checkpoint and $T = 1.0$. Panels A and B use only a species token and the upstream marker (no CDS); panels C and D use CDS prompts. **(A)** Natural versus generated 6-mer frequencies under matched and mismatched species tokens (673 *E. coli* and 441 *B. subtilis* upstreams; 20 outputs each); Pearson r is reported. **(B)** Positionwise natural (dashed) and generated (solid) base profiles near the Shine–Dalgarno region; shading is ± 1 standard deviation. **(C)** Base profiles under the native and shuffled CDS prompts, and their difference. **(D)** Per-gene GC content of natural upstreams and outputs generated from the native CDS prompt.

The generator captures broad sequence patterns. Figure 3A compares the frequency of each of the 4096 possible 6-mers in generated and natural upstreams. Prompted with its own species token, the model reproduces natural 6-mer frequencies closely ($r = 0.896$ for *E. coli*, $r = 0.938$ for *B. subtilis*) compared with the other species' token the correlation falls to 0.774 and 0.770, so the token carries species-level information. This measure is position-blind, as a high 6-mer correlation is achievable from composition alone and does not by itself indicate regulatory grammar.

The model captures the Shine-Dalgarno (SD) sequence. Figure 3B shows base probabilities over the final 20 bp before the start codon for natural (dashed) and generated (solid) upstreams. The generated profiles track the natural ones in both species, including the G peak near position -10, and capture the species-specific form: in real sequences the peak reaches 0.63 in *B. subtilis* against 0.54 in *E. coli*, and the generated profiles give 0.63 against 0.49. However, this feature is the easiest to capture, since the SD sequence sits at nearly the same offset in every gene (Fig. D.1B).

Changing the conditioning CDS has almost no effect on generation. In Figure 3C we prompt the model with the real CDS and with a shuffled version of that same CDS, which keeps its base composition but destroys the gene it encodes. If the model were reading the gene, the two prompts should lead it to write different upstreams. The first two tracks give the probability of each base across the generated 200 bp and are almost indistinguishable. The third track subtracts one from the other and stays flat at zero, mean $|\Delta p| = 0.003$. Therefore, the CDS is present in the prompt but

effectively absent from the output. Table A.1 puts the comparison in bits of KL divergence, where zero means the CDS made no difference to what the model wrote. Even deleting the CDS from the prompt entirely costs only 2.8 bits over 200 positions, or 0.014 bits per position. This is small rather than zero: a base distribution can carry up to 2 bits, so the most violent perturbation available moves under 1% of what could move. The response is also in the wrong place. Fig. A.1 resolves it by position: it peaks in the first 50 bases, which the model writes immediately after reading the CDS, and decays with distance, while the Shine–Dalgarno and start-codon region carries less than its share of the window. The model therefore writes much the same upstream whatever gene it is given, and what little it does condition on the CDS is spent away from the positions that carry regulatory function.

Generated upstreams have little gene-specific variation. Figure 3D compares the GC content of natural and generated upstreams, with one value for each gene. Natural upstreams vary substantially between genes, with standard deviations of 6.6% in *E. coli* and 5.4% in *B. subtilis*. Generated upstreams show less variation, with between-gene standard deviations of only 2.3% and 2.4%. Their GC values concentrate in the high-density region of the corresponding natural distribution but span a narrower range. The model therefore reproduces typical species-level composition while underrepresenting the natural diversity among genes.

5 Contrastive loss recovers pairwise signal

Figure 1A shows a relationship between a gene and the upstream region. C3P model is introduced to assess if this pairwise signal can be captured by a model. C3P provides such a reference: its 100M-parameter promoter encoder was contrastively aligned to a frozen protein encoder on 88 million bacterial pairs, increasing similarity for native promoter–protein pairs relative to mismatches. This pairwise loss directly targets CDS–upstream association. C3P acts as a retrieval model, not a generator; we use it only to test whether sequence contains a learnable pairing signal. We test on *E. coli*, a species that was excluded from the training set of C3P [5].

For each of the 500 singleton *E. coli* genes in the retrieval set, we rank the true upstream against 99 same-genome decoys over 20 resamples, scoring each autoregressive model in two ways: candidate-centred likelihood and embedding cosine similarity (Appendix A). Likelihood performs better; Evo 2 reaches 6.0% top-1 by likelihood but 1.0% by embedding.

C3P ranks the correct upstream first for 82.7% of genes (median rank 1), providing a positive reference that the pairing is recoverable under a different objective. Autoregressive top-1 accuracy remains low: Evo 1 6.3%, Evo 2 6.0%, fine-tuned Evo 2.8% and HyenaDNA 1.0% (median ranks 25, 31, 36 and 47, respectively). All of these use the candidate-centred scorer, which subtracts each candidate's own unconditional likelihood so that a generically likely upstream cannot win on composition alone. Removing that correction raises Evo 2 from 6.0% to 7.5%—the highest autoregressive value we measured, and still an order of magnitude below C3P (Appendix A).

6 Failure analysis, boundary conditions and lessons

Next-token cross-entropy is a good objective function when selection constrains most of the sequence. In a protein, nearly every residue is under selection—for fold, stability or activity—so a model that predicts the next residue better is also a model that has learned more about the protein, and the two objectives cannot be separated. This is why the same loss yields protein models that recover structure and fitness [9, 10, 14]. However, a bacterial upstream contains a small number of identifiable elements including distal and proximal promoter boxes, the SD sequence, and transcription-factor binding sites—embedded within spacer sequence that makes up most of the region. These elements are also diverse and evolutionarily flexible. This motivates a hypothesis, which our measurements are consistent with but do not establish: because next-token cross-entropy is summed over every position, most of the training signal comes from the largely unconstrained background rather than from the short identifiable elements, so lower-order properties such as GC content and k-mer composition offer an easier way to reduce the loss than motif identity and spacing. We state this as a hypothesis because none of our experiments manipulate the objective. Testing it requires an objective-level ablation—for instance re-weighting the loss at annotated element positions, holding data and architecture fixed—which we do not perform here.

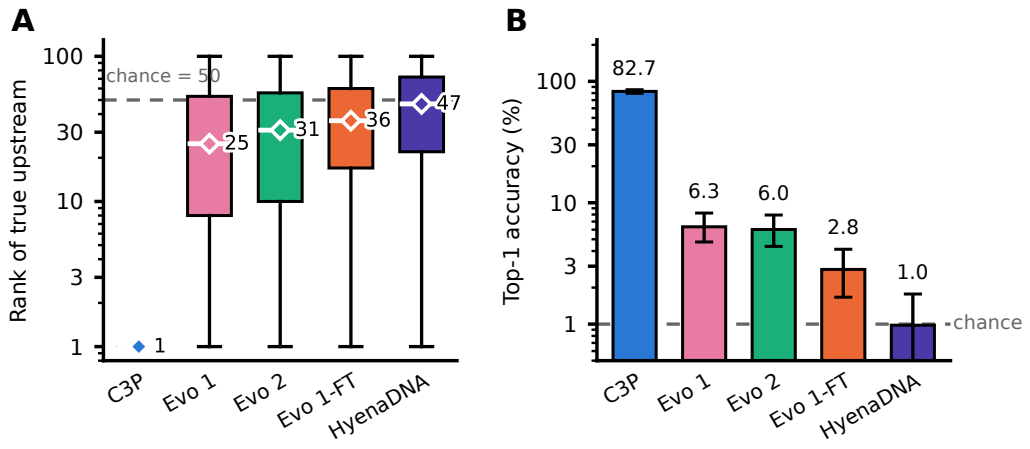


Figure 4: **Retrieval of matched CDS–upstream pairs.** For 500 *E. coli* genes, the true 200-bp upstream was ranked among 99 same-genome decoys over 20 resamples. C3P uses protein–upstream cosine similarity; autoregressive models use candidate-centred conditional log likelihood. **(A)** Rank distributions (diamond, median; box, IQR; whiskers, 1.5×IQR). **(B)** Top-1 accuracy with 95% bootstrap intervals; the dashed line marks chance. Two CDSs longer than 4.3 kb were excluded as queries but retained as decoys.

What the models write are flat profiles of AT-rich elements for the regulatory sequence. Across the 41 bp that span the proximal box the probability of each base changes slightly with position (Fig. 2B): real promoters dip to 1.72 bits of entropy inside the box, while no model moves by more than 0.03 bits. The level they settle at is close to correct, 58–60% AT against 61.5% in real upstreams, and this is already the composition of their training data. The non-coding sequence of all three corpora is AT-rich (50.8%, 54.7% and 59.2% AT for Evo 1, Evo 2 and HyenaDNA; Fig. E.1), and HyenaDNA arrives at the same output composition having seen only human sequence. The same measurement on coding sequence gives the opposite result: prompted at the midpoint of a CDS, these models write a profile that varies from position to position and follows a similar profile to natural CDS sequences (Appendix B). Most bases in a CDS are constrained by the genetic code, whereas identifiable regulatory elements occupy only about 20 of the 200 upstream bases. This contrast is what the hypothesis above would predict: patterns present at most positions are learned more readily than sparse, short regulatory grammar [1, 15]. It is also consistent with two accounts we cannot separate from it here. First, free-running decoding compounds error—once a spacer drifts from its canonical length nothing restores it—which is a property of sampling rather than of the training objective. Second, uniform position weighting is shared by masked genomic language models, so these results may reflect a limit of likelihood pretraining on this class of sequence rather than of next-token prediction specifically.

Among the evaluated models, Evo 2 is the only best-performing model that generates real CDS profiles in the coding-sequence control (Appendix B). Evo 2's training procedure may contribute to this difference. Its initial short-context training upweights genic windows, after which midtraining uses longer sequences and a greater proportion of whole genomes [2]. This emphasis on gene-rich sequence may help explain Evo 2's stronger performance in the CDS control. More generally, the result supports that a model learns most reliably the sequence properties emphasized during training.

Where this account is incomplete. PromoGen2 is the natural exception. It was trained on curated promoter sequence, where regulatory elements are not sparse, so the hypothesis above predicts that it should succeed. It does not: it places the σ^{70} proximal box in 10.8% of generations and recovers the Shine–Dalgarno in 12.4%, both below its own scrambled control (12.3% and 14.5%; Table A.2). Either the sparsity account is incomplete, or this checkpoint is being run outside its intended use as a conditional design model. Two further limits bound what we can claim. Evo 2 appears at a single scale (1B), so the monotonic rise of its placement accuracy with context (Fig. 2D) cannot distinguish an objective that does not learn positional grammar from checkpoints that did

not. And fine-tuning *lowered* retrieval, from 6.3% for base Evo 1 to 2.8%; Fig. 3D indicates why, as the between-gene standard deviation of GC content falls from 6.6% to 2.3%, making generated upstreams more species-typical and less gene-specific.

What these results do not show. Everything here is measured on sequence. We show that a gene and its upstream carry matching statistical signal, and that some models can pick that signal up. We do not show that a gene causes its promoter to look a particular way, that a retrieved pair would express well, or that any generated upstream would work in a cell.

The obvious next model is a hybrid. The two objectives fail and succeed in complementary places. Cross-entropy writes sequence but is dominated by composition; the contrastive loss cannot write anything, yet it is the only objective here that ties a gene to its own upstream (82.7% against 6.3% under the primary scoring rule and 7.5% under the most favourable variant, Fig. 4). The direct test is to train one model under both at once: the same next-token cross-entropy, unchanged, plus a contrastive term that scores each gene against its own upstream and against the upstreams of other genes in the same genome. Cross-entropy keeps the ability to write sequence; the contrastive term adds the gene–upstream pairing that cross-entropy does not prioritise. We say "does not prioritise" rather than "cannot represent": autoregressive retrieval is above chance and the fine-tune did acquire a Shine–Dalgarno signal, so the pairing is partly captured, just weakly. The next thing to vary is the objective function, not the architecture.

Reproducibility statement

All benchmark inputs derive from the public reference genomes and annotations identified above. Code and data are released at `https://github.com/MarcAmil30/composition-without-grammar`: filtered gene lists, prompt manifests, checkpoint identifiers, evaluation scripts, per-model framing, the summary tables behind every figure, and the figure-generation code, with a notebook that redraws each figure in paper order. The raw generations, likelihood matrices and per-shard probability tensors are tens of gigabytes and are not redistributed; neither is the CDS-conditioned Evo fine-tune. The repository states which scripts run from the released tables and which require that material. The Evo fine-tune starts from `evo-1-8k-base` and uses learning rate 10^{-4}, batch size 2, at most $80{,}000$ updates, early-stopping patience 2, and no label mask. Training uses the designated `TRAIN` file; a separate `TEST` file is used for the loss audit, but the generation and autoregressive-retrieval benchmarks were not filtered against the training genes or their homologues. Generations use 20 samples at $T = 1.0$ with no top-k or nucleus truncation (seed 0); CDS perturbations use seed 20260801; and retrieval uses seed 0 with 20 decoy sets per gene. The public C3P checkpoint is a transformer promoter encoder aligned in 256 dimensions to a frozen ESM-2 (150M) protein encoder; it was trained for 10 epochs on 88 million pairs from $\sim$23,000 species. Its species-held-out split, including exclusion of *E. coli*, is documented by Dufault et al. [5]; the paper does not report homology-disjoint filtering.

Ethics statement

This study uses public bacterial reference genomes and annotations; it contains no human or personally identifiable data. No sequence was synthesized and no organism was modified. We report model limitations and sequence-level evaluations without claiming biological function.

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Appendix guide

Appendix A defines the metrics and reports detailed values; Appendix B gives the coding-sequence positive control. Appendices C–F document, in order, the Fig. 1A analysis, upstream-only benchmark, training corpora and CDS-prompted benchmark. None is required to follow the main argument.

A Evaluation metrics and detailed results

The metrics below address four questions: *where* a model places an element, *which bases* it writes there, how much its output *changes* with the prompted CDS, and how well it *ranks* the true upstream.

A.1 Definitions and interpretation

Base profile. Let G be the gene set, $N = 20$ the generations per gene, $s_{g,n}[i]$ the base at position i in generation n for gene g, and $b \in \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$. The indicator $\mathbf{1}[\cdot]$ is 1 when its statement is true and 0 otherwise:

$$p_g(i,b) = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}[s_{g,n}[i] = b]\,, \qquad \bar{p}(i,b) = \frac{1}{|G|}\sum_{g\in G} p_g(i,b), \tag{A.1}$$

Thus, $p_g(i,b)$ is the fraction of a gene's generations with base b at position i, and $\bar{p}(i,b)$ averages this fraction over genes. The entropy $H(i) = -\sum_b \bar{p}(i,b)\log_2 \bar{p}(i,b)$ is 2 bits when the four bases are equally common and 0 when all sequences have the same base; a dip therefore shows a preferred base. Only Eq. A.4 uses predicted probabilities rather than sampled output.

6-mer correlation. For each sequence set, we count the 4096 possible six-base words and convert the counts to frequencies. Pearson's r compares the natural and generated frequency vectors: $r = 1$ means their relative frequencies match, whereas $r = 0$ means no linear agreement. This metric ignores where each 6-mer occurs.

Placement. For each sequence $s_{g,n}$, we scan a window of length L for a promoter box of width $w = 6$. Here, $M(k,b)$ is the probability of base b at motif position k in annotated promoters of the same sigma factor, $q(b)$ is its background probability in upstream DNA, and $J = L - w + 1$ is the number of possible offsets. The log ratio rewards promoter-enriched bases; its sum over the six positions scores each offset:

$$\begin{aligned} \hat{\jmath}_{g,n} &= \arg\max_{1\le j\le J} \sum_{k=1}^{w} \log_2 \frac{M(k,\, s_{g,n}[j{+}k{-}1])}{q(s_{g,n}[j{+}k{-}1])}, \\ \mathrm{Acc}_\tau &= \frac{1}{|G|N}\sum_{g,n} \mathbf{1}\left[\left|\hat{\jmath}_{g,n} - j^*_g\right| \le \tau\right]. \end{aligned} \tag{A.2}$$

Here, $\hat{\jmath}_{g,n}$ is the best-scoring offset, j^*_g the annotated offset, and τ the allowed error. Accuracy is the fraction of generations placed within τ bases of the annotation. With $\tau = 2$ and $J = 40$, uniform placement gives $(2\tau+1)/J = 12.5\%$. Because the scan always selects a best offset, this metric measures position rather than motif presence.

Motif identity. Eq. A.2 gives the start $\hat{\jmath}_{g,n}$ of the highest-scoring six-base window. We count how many bases in that window match gene g's annotated hexamer, then average over its 20 generations and over genes. A perfect score is $6/6$, although the scan gives real sequences $4.99/6$ for the proximal box and $5.01/6$ for the Shine–Dalgarno because it sometimes selects another window. The negative control shuffles each generation before repeating the scan.

CDS sensitivity. Let C_0 be the native CDS and C_k a perturbed version. The superscript on $\bar{p}^{\,C}(i,b)$ indicates which CDS prompted the generated sequences. We first compare the two averaged base profiles:

$$\overline{|\Delta p|} = \frac{1}{200\cdot 4}\sum_{i=1}^{200}\sum_{b} \left|\bar{p}^{\,C_0}(i,b) - \bar{p}^{\,C_k}(i,b)\right|, \tag{A.3}$$

where i is one of 200 upstream positions and b one of four bases. The absolute differences are averaged over all 800 position–base combinations, so zero means the two population profiles are identical. This is the quantity in Fig. 3C.

The second measure compares prompts separately for each gene. Here $p_g^C(i, b)$ is the model's predicted probability of base b for gene g at position i under CDS prompt C. Both prompts are evaluated against the same natural upstream by teacher forcing, so only the CDS differs:

$$\mathrm{KL}(C_0 \| C_k) = \left\langle \sum_{i=1}^{200} \sum_{b} p_g^{C_0}(i, b) \, \log_2 \frac{p_g^{C_0}(i, b)}{p_g^{C_k}(i, b)} \right\rangle_g . \tag{A.4}$$

The sums compare the two predicted base distributions at every position, and $\langle \cdot \rangle_g$ averages the result over genes. KL is zero only when both prompts give identical predictions; larger values indicate greater CDS sensitivity. It is reported over all 200 positions, with division by 200 giving bits per position. Unlike Eq. A.3, this within-gene comparison can detect changes that cancel in the population average. Two limits of its scope should be stated. The model reads the *natural* upstream at every step rather than its own output, so Eq. A.4 measures how far the model's predictions move when the CDS changes, not how far its generated sequences would move; Fig. 3C is the generation-level counterpart. And a 200-position sum says nothing about where the movement sits, which Fig. A.1 reports separately. Fig. 3C shows the shuffled-CDS profile comparison; Table A.1 reports the gene-level KL results.

Retrieval. Query i is a CDS and candidate j is a possible upstream.

Framing, per checkpoint. Each model is given the arrangement it was trained on; presenting a model an arrangement it never saw returns finite but meaningless likelihoods. Sequences are written $5' \to 3'$ and every tokenizer was verified to emit exactly one token per nucleotide.

- *Fine-tuned Evo (PROK).* Trained CDS-first with segment tokens, so the input is `+#<CDS>+$<upstream>`, where `+` is the species token and `#` and `$` open the CDS and the $5'$ region. The scored tokens are the 200 upstream bases, giving $\log P(\text{upstream} \mid \text{CDS})$.
- *Evo 1 (8k), Evo 2 (1B), HyenaDNA.* These are base checkpoints trained on bare genomic DNA with no delimiters, so the input is the natural order `<upstream><CDS>` with no separator. A causal model in this order cannot let the upstream attend to the CDS, so the conditional is necessarily taken in the other direction: the scored tokens are the CDS, giving $\log P(\text{CDS} \mid \text{upstream})$. Retrieval still ranks upstreams, because varying the candidate upstream varies the context the CDS is scored in.
- *C3P.* No likelihood head, so cosine in its trained space is its score. The query is the *protein* rather than the CDS, since that is the modality it was aligned with; the candidate is the 200 bp upstream, taken as the $3'$-most 200 bases. Both are projected to 256 dimensions and L_2-normalised; the protein vector is the attention-masked mean of the ESM-2 final hidden states.

Query counts. Evo 2, HyenaDNA and C3P score all 500 queries. For the two StripedHyena checkpoints (Evo 1 and the fine-tuned model) the two longest CDSs (7,020 and 4,029 bp) exceed the memory budget of a single forward pass; they are dropped as *queries* and retained as *candidates*, leaving 498 queries against the same 100-way candidate sets.

Likelihood score. For model m, $\ell_{ij}^{(m)}$ is the mean per-base log likelihood of the scored segment named above when query i is paired with candidate j. Some candidates score highly with almost every CDS, so we remove this general advantage:

$$s_{ij}^{(m)} = \ell_{ij}^{(m)} - \frac{1}{|G|} \sum_{i' \in G} \ell_{i'j}^{(m)}, \tag{A.5}$$

where i' runs over all CDS queries in gene set G. The second term is candidate j's average score across those queries; subtracting it leaves $s_{ij}^{(m)}$ high only when j fits query i unusually well.

Embedding score. An autoregressive model converts the CDS and each upstream into separate vectors by averaging its final hidden states across sequence positions. After centring and normalizing the vectors, we rank upstreams by cosine similarity: a higher value means that the model represents

the CDS and upstream more similarly. C3P uses the same basic ranking idea but replaces the CDS with its protein. Its contrastive training places protein and upstream vectors in a shared space, so the upstream with the highest cosine similarity is ranked first without candidate correction.

For both approaches, each query is ranked against its true upstream and 99 same-genome decoys over 20 seeded resamples; chance is 1% top-1 and median rank 50. The scoring-variant sweep was run for Evo 2 and HyenaDNA; Evo 1 and the fine-tuned model were evaluated under the candidate-centred scorer only. Across the swept variants the best autoregressive result is 7.5% (Evo 2 without the candidate correction), against C3P's 82.7%. Confidence intervals bootstrap genes, not the correlated resamples.

A.2 Detailed results

Table A.1 reports CDS sensitivity under each perturbation; Table A.2 reports promoter and Shine–Dalgarno placement by model; Fig. A.2 gives the corresponding Shine–Dalgarno identity scores.

Table A.1: **Changing the CDS has little effect on upstream predictions.** $\mathrm{KL}(C_0 \,\|\, C_i)$ is summed over 200 positions while teacher-forcing the same natural upstream, so only the CDS changes. Zero means identical base probabilities; removing the CDS gives 2.8 bits in total, or 0.014 bits per position.

	CDS given to the model	*E. coli*	*B. subtilis*
C1	synonymous codons permuted (protein, counts, GC fixed)	0.49	0.58
C2	codon order shuffled	1.05	1.14
C3	substituted from another gene	2.05	2.17
C4	removed entirely	2.80	2.81
C5	first 30 codons shuffled	0.02	0.03
C6	codons 31–end shuffled	0.96	1.04

Table A.2: **Promoter and Shine–Dalgarno placement by model.** Percentage of generations whose best PWM match lies within ±2 bp of the element's annotation, by sigma factor. Bold marks the best model per column; italic rows are natural sequence and shuffled model output. No model exceeds the scrambled control by more than a few points in any column, so the failure is not confined to σ^{70}.

	proximal box, by sigma factor					Shine–Dalgarno
Model	σ^{70}	σ^{38}	σ^{32}	σ^{24}	σ^{54}	all
real sequence	*67.6*	*63.4*	*87.8*	*68.8*	*82.8*	*78.6*
Evo 1 (8k)	10.5	11.9	13.6	11.0	12.6	**15.2**
Evo 1 (131k)	10.7	10.9	**17.3**	**16.1**	15.1	13.8
Evo 2	**13.7**	11.9	11.0	12.2	**19.5**	14.7
HyenaDNA	11.6	11.7	10.8	11.2	12.0	13.6
PromoGen2	10.8	**12.5**	15.3	11.2	13.3	12.4
scrambled control	*12.3*	*12.2*	*12.2*	*12.8*	*13.4*	*14.5*

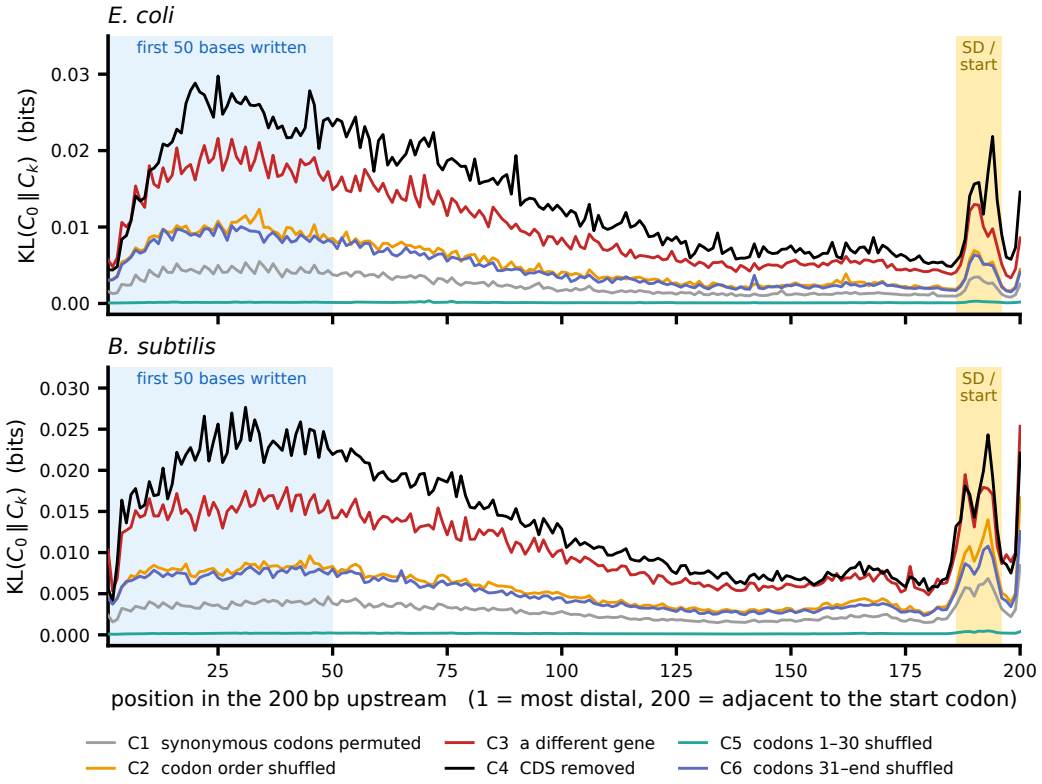


Figure A.1: **Where the CDS sensitivity sits along the upstream.** *How this is assessed.* For one gene we give the model its real CDS and then walk along the real 200 bp upstream one base at a time, recording the four base probabilities the model predicts at each position. We repeat this with the CDS perturbed (C1–C6, Table F.1) and the upstream unchanged, and plot the per-position KL divergence between the two sets of predictions (the summand of Eq. A.4, before the sum over positions). A larger value at a position means the model's expectation for that base depends on which gene follows. Position 1 is the most distal base and the first one the model writes; position 200 is adjacent to the start codon. *What it shows.* The response is a decay from the prompt boundary, not a feature of the regulatory elements. The first 50 positions hold 38–43% of the total in *E. coli* against a 25% share of the window (1.5–1.7×), whereas the Shine–Dalgarno and start-codon region (shaded, positions 186–196) holds 0.84–0.94× its share for every large perturbation. The exception is C5, which shuffles only codons 1–30, the stretch adjacent to the Shine–Dalgarno: it is the smallest effect overall (0.02 bits) yet the only condition enriched at that region (1.6× in *E. coli*, 2.1× in *B. subtilis*). Per-condition totals, peaks and enrichments are released in `results/supp_kl_perposition_stats.csv`. Curves are averaged over genes; the spread across individual genes is not resolved here.

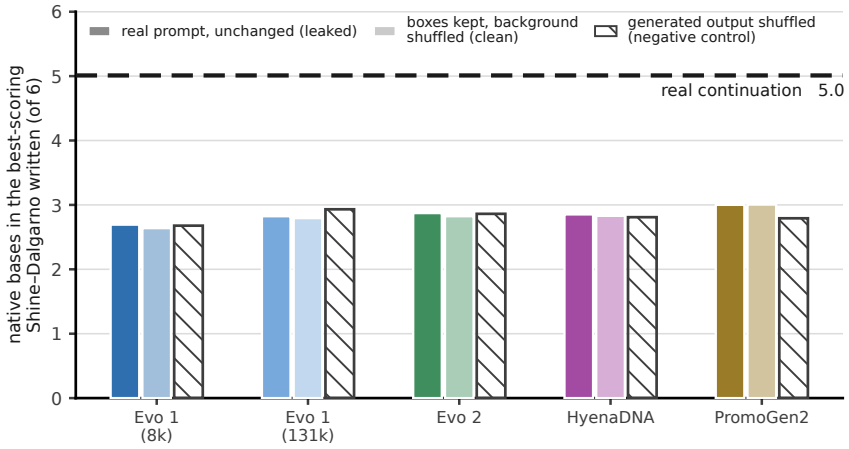


Figure A.2: **The best Shine–Dalgarno a model writes is not the right one.** Fig. 2C repeated for the Shine–Dalgarno task on all 681 genes: the highest-scoring SD PWM window in each generation, scored by how many of its six bases match that gene's annotated SD core. Bars are the leaked and clean prompt conditions and the shuffled-output negative control. The dashed line is the same scan on the real continuation ($5.01/6$). Every model scores 2.6–3.0, and none exceeds its own shuffled output by more than 0.24 bases.

B Coding sequence as a positive control

The same models were asked to continue coding sequence instead of regulatory sequence, under the same free-running decode. Figure B.1 shows the positionwise profiles, and Fig. B.2 compares their distance from real coding and upstream sequence.

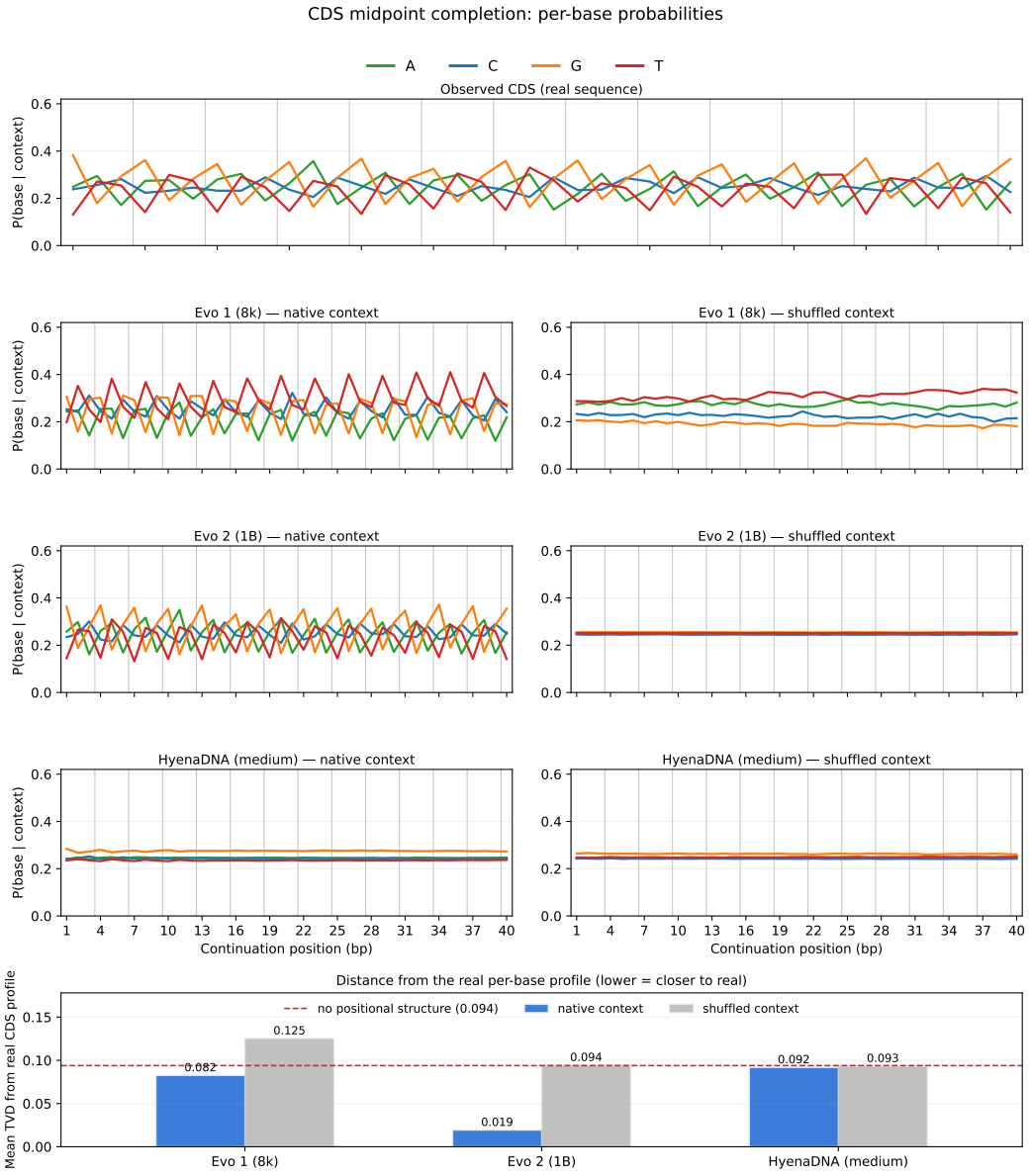


Figure B.1: **CDS midpoint completion: per-base probabilities.** Each of 672 *E. coli* genes was cut at the midpoint of its coding sequence and continued for 40 bp (1 greedy plus 20 sampled decodes per gene and condition). **Top:** probability of A, C, G and T at each continuation position in the real coding sequence. **Middle:** the same profiles for Evo 1 (8k), Evo 2 (1B) and HyenaDNA, prompted with the native context (left) or with a shuffled context (right). **Bottom:** mean total variation distance between each model's profile and the real one; 0 means identical profiles. The dashed line (0.094) is the distance obtained by using the same overall base composition at every position.

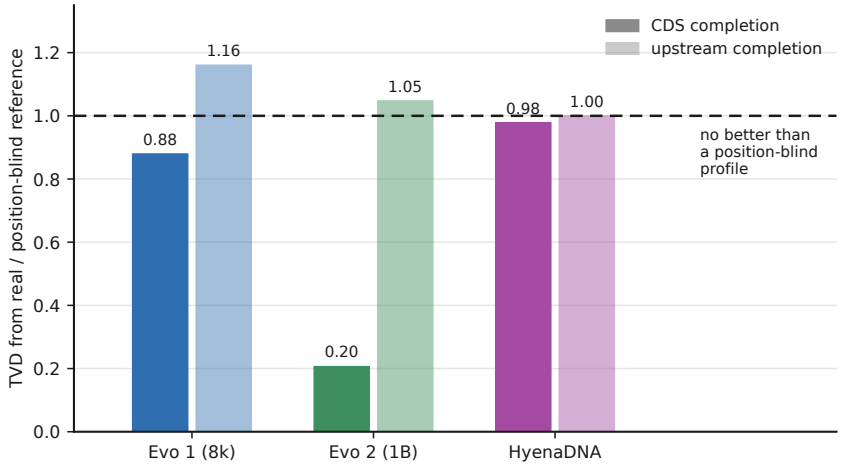


Figure B.2: **The same models are further from real upstream sequence than from real coding sequence.** Mean total variation distance between each model's per-base profile and the real one, $\mathrm{TVD}(i) = \frac{1}{2}\sum_b |p_{\mathrm{model}}(i,b) - p_{\mathrm{real}}(i,b)|$, where $p(i,b)$ is the frequency of base b at position i. TVD is 0 when the generated and real profiles match. We divide it by the value obtained when the same overall base composition is used at every position (0.094 coding, 0.067 upstream; dashed line). Below 1 means positional structure was captured, 1 composition alone, above 1 that the composition is off as well; finite-sample noise alone gives 0.026 and 0.031. CDS completion: 672 *E. coli* genes cut at the CDS midpoint and continued for 40 bp. Upstream completion: 487 σ^{70} promoters over the 41 positions following the distal box (the Fig. 2B window). PromoGen2 and Evo 1 (131k) were not run on both windows and are omitted.

C The gene–upstream association of Fig. 1A

Pairs and controls. We paired 727 *E. coli* K-12 singletons with singleton orthologues from MMseqs2 Gammaproteobacterial families, requiring $\geq 30\%$ BLASTP identity, $\geq 70\%$ coverage and a full 200 bp upstream. This gave 22,393 pairs from 408 genomes. Controls were unrelated singletons from the same target genome, length-matched to $\pm 10\%$, thereby matching genome composition and phylogenetic distance while removing homology.

Why conserved gene order does not drive this panel. Both filters above are applied to every gene on both sides of every pair, and together they rule out the obvious synteny confound. "Singleton" means an operon size of one, so the gene is not co-transcribed with a neighbour; "a full 200 bp upstream" means the annotated distance to the nearest upstream feature is at least 200 bp, so the window stops before the neighbouring gene and contains no neighbouring coding sequence. A window here is therefore intergenic by construction, and conserved gene order cannot produce upstream similarity by placing a conserved CDS inside it. The controls inherit the same two filters, so test and control pairs are matched on gene architecture as well as on genome composition, phylogenetic distance and upstream length. The coverage figures quoted in Appendix F (41% and 32% of windows clear of the nearest upstream feature) describe the CDS-prompted benchmark, which applies no such distance requirement, and do not apply to this panel.

One weaker version of the concern does survive. Clearing the neighbouring *gene* does not clear shared regulatory architecture: where a gene and its orthologue both lie downstream of a divergently transcribed neighbour, the conserved sequence in the window may be that neighbour's promoter rather than the focal gene's. This remains regulatory sequence, but it is not necessarily evidence of coevolution with the focal CDS.

Upstream similarity (D_2^*). A direct 4-mer comparison can be misleading because two AT-rich sequences share many short words even when they are unrelated. We therefore compare observed counts with those expected from each sequence's own background composition. For a 4-mer w, X_w is its observed count and $\mathbb{E}[X_w]$ its expected count. The expectation uses an order-2 Markov model, which predicts each base from the previous two bases: $\mathbb{E}[X_w] = N(w_{1:3})N(w_{2:4})/N(w_{2:3})$, where $N(u)$ counts substring u and the subscripts select the first three, last three and middle two bases of w. We then calculate

$$\tilde{X}_w = \frac{X_w - \mathbb{E}[X_w]}{\sqrt{\mathbb{E}[X_w]}}, \qquad D_2^* = \frac{\sum_w \tilde{X}_w \tilde{Y}_w}{\sqrt{\sum_w \tilde{X}_w^2}\,\sqrt{\sum_w \tilde{Y}_w^2}}, \tag{C.1}$$

Here, $\tilde{X}_w$ is positive when w occurs more often than expected and negative when it occurs less often; $\tilde{Y}_w$ is the same value for the second sequence. D_2^* compares these corrected patterns across all 256 4-mers: 1 means the patterns match and 0 means no agreement. It therefore measures similarity beyond that explained by background base composition.

Third-position identity. A codon contains three bases, and changes at its third base often do not change the encoded amino acid. For each aligned gene pair, we therefore calculate the fraction of codons with the same third base, using pairs with at least 30 aligned codons. A high value indicates more closely related coding sequences and a low value greater evolutionary divergence. This is the x axis in Fig. 1A and measures relatedness, not gene function.

D The upstream-only benchmark

Not all five sigma factors build promoters from $-35/-10$ elements (σ^{54} uses $-24/-12$), so we write *distal* and *proximal* box throughout. Figure D.1 summarizes the dataset and controls, Fig. D.2 shows the box motifs, and Fig. D.3 shows the spacer-length distributions.



Figure D.1: **Composition and controls of the upstream-only benchmark.** 681 RegulonDB-annotated *E. coli* genes with up to 200 bp before the start codon. **(A)** Sigma-factor counts; promoter-box analyses use the 671 genes in the five displayed factors, while one σ^{28} and nine unassigned genes remain in Shine–Dalgarno analyses. **(B)** Annotated element midpoints relative to the start codon. **(C)** Distal-to-proximal box spacing. **(D)** Information logo for the final 20 bp. **(E)** Fraction of each clean prompt held fixed as an annotated element. **(F)** Natural and clean-prompt GC content. The Shine–Dalgarno emerges de novo in (D) with no motif assumed, and the clean background holds every annotated element at its real position (E) while preserving mononucleotide composition exactly (F), so a score against it cannot be explained by composition.

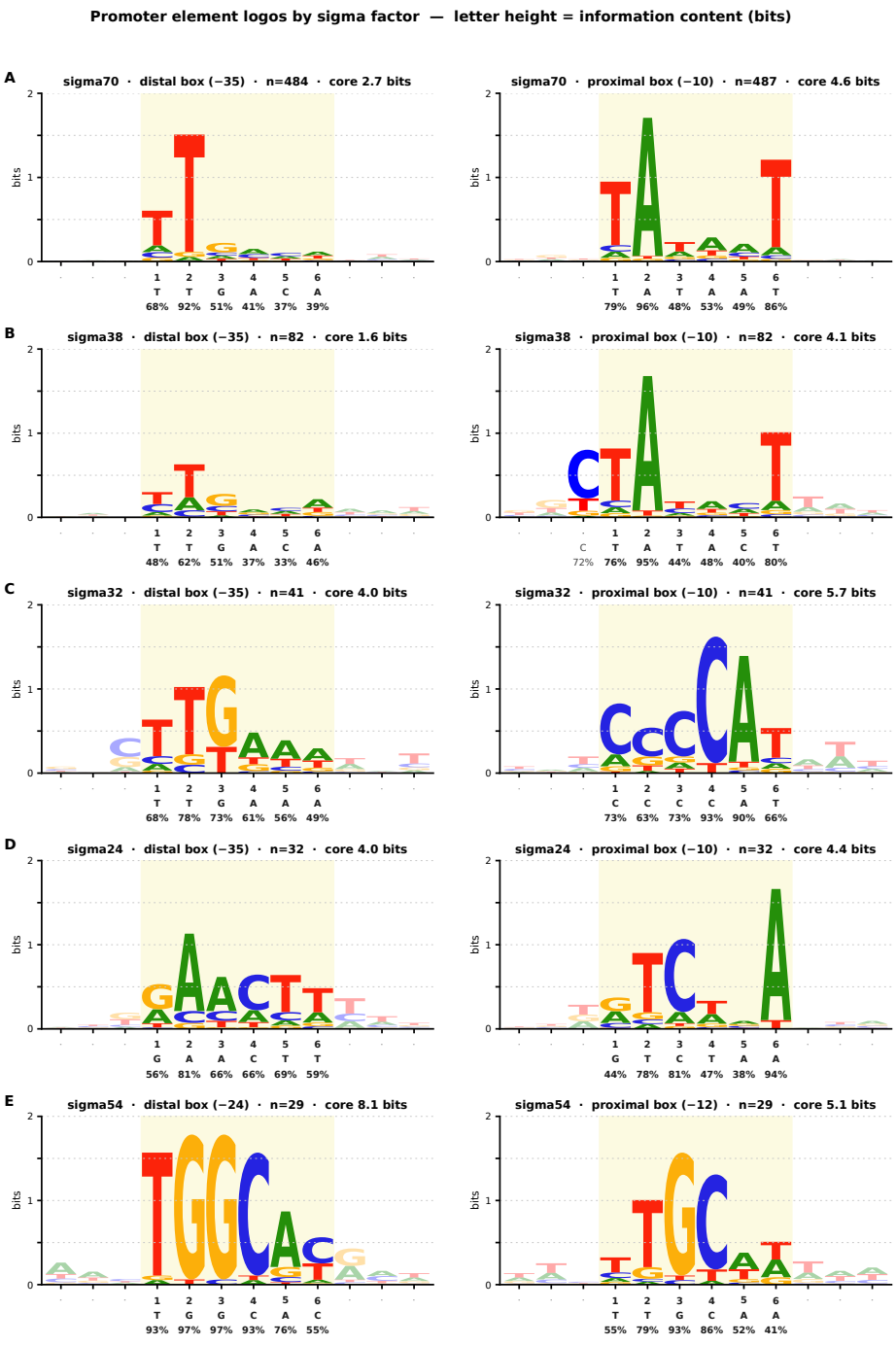


Figure D.2: **Promoter-box logos by sigma factor.** Information logos for distal and proximal boxes in RegulonDB-annotated *E. coli* upstreams ($n \geq 25$ per factor). Six-base cores are shaded and three-base flanks are faded; the modal base and frequency are printed below each core position. Windows are anchored at the distal box's $5'$ end and the proximal box's $3'$ end; the benchmark reproduces the published consensus for every factor.

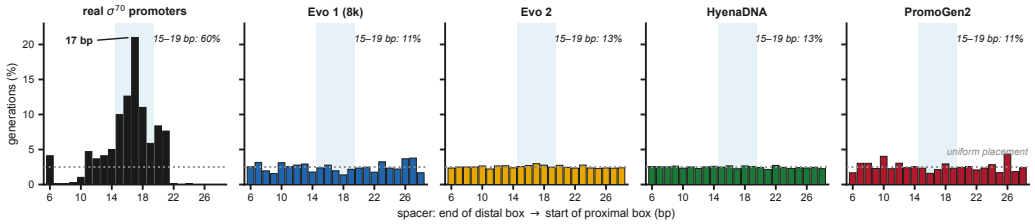


Figure D.3: **The spacer distribution, one histogram per model.** Each model's best-scoring proximal-box position, read as a spacer from the distal box, on a shared y-axis. Real σ^{70} promoters concentrate at 17 bp; every model spreads the spacer almost uniformly. Shading marks the canonical 15–19 bp window, the dotted line uniform placement (2.5% per offset), and the inset number the fraction inside the canonical window. Real promoters fall in the canonical window 60.5% of the time and at exactly 17 bp 21.0% of the time; the models give 10.7–13.5% and 2.0–3.0%, against 12.5% and 2.5% for uniform placement. Presence is not the failing quantity: the scan finds a box in 93–97% of generations and in 95–97% of shuffles of those same generations, because an AT-rich stretch supplies a proximal-box-like hexamer by chance.

E What the models were trained on

Non-coding bases—those outside predicted coding regions—are an upper bound on what a model could have learned promoters from.

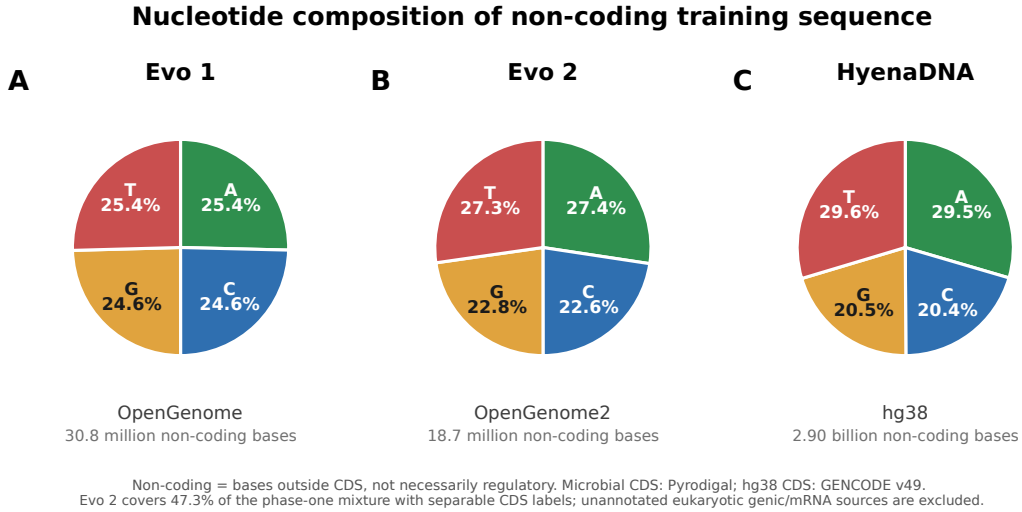


Figure E.1: **Nucleotide composition of the non-coding sequence in each model's training corpus.** Bases outside predicted CDS intervals, sampled reproducibly from each released corpus. **(A)** Evo 1 (OpenGenome). **(B)** Evo 2 (OpenGenome2), restricted to the 47.3% of the phase-one mixture whose records carry separable CDS labels; unannotated eukaryotic genic and mRNA sources are excluded. **(C)** HyenaDNA (hg38). PromoGen2 is omitted: it is trained on curated promoter sequence, not whole genomes. The corpora differ (50.8%, 54.7% and 59.2% AT) but the outputs do not (60.4%, 57.7% and 59.7%, against 61.5% in real *E. coli* upstreams), so matching the composition of a bacterial upstream is a low bar.

F Composition of the CDS-prompted benchmark

The benchmark contains protein-coding, in-frame singletons of at least 30 codons with a complete 200 bp upstream: 1,885 from *E. coli* and 1,272 from *B. subtilis*. Only 41% and 32% of these windows are clear of the nearest upstream feature, so overlap is a stratifier; restricting the analysis to singletons limits inference to independently regulated genes. This coverage figure applies to the CDS-prompted benchmark only. The Fig. 1A pairs additionally require at least 200 bp to the nearest upstream feature (Appendix C), so every window in that panel is clear.

The seven conditions are defined in Table F.1. C1 and C2 rearrange codons, C3 substitutes another gene and C4 removes the CDS. C5 shuffles the first 30 codons and C6 the remainder, contrasting the region nearest the ribosome-binding site with the region nearest the generation point. Cross-gene upstream identity is near the composition floor (0.255 and 0.269), so a species-generic upstream is not a gene-specific match.

Table F.1: **The seven CDS perturbation conditions.** Percentages are mean nucleotide changes in *E. coli*/*B. subtilis*. The start codon is fixed; shuffled conditions preserve length, codon counts and GC, while C3 is length-matched and GC-matched in the mean. KL results are in Table A.1; C2 is shown in Fig. 3C.

	CDS given to the model	nt changed (E/B)	CDS GC vs. C0
C0	the real CDS (reference)	—	—
C1	synonymous codons permuted, protein unchanged	$20.0/20.6\%$	identical
C2	codon order shuffled	$72.0/71.2\%$	identical
C3	a different, length-matched gene's CDS	$71.9/71.9\%$	matched in the mean
C4	removed entirely	—	—
C5	codons 1–30 shuffled; remainder fixed	$10.7/11.9\%$	identical
C6	codons 31–end shuffled; first 30 fixed	$60.8/58.7\%$	identical